

FMUSER FUQ-4443 IP QAM Modulator 1024 IP Inputs 32 RF Outputs GE Ports with SFP DVB-C ITU-T J.83 A-B Multiplexing Scrambling Headend Broadcast System Product Overview



The FMUSER FUQ-4443 is an IP QAM modulator with 32-channel multiplexing, scrambling, and modulation, supporting up to 1024 IP inputs via GE ports at max 840Mbps each and outputting 32 non-adjacent carriers through dual RF ports (50-960MHz). It processes signals using PID remapping, PCR adjusting, and PSI/SI editing for optimization. This unit features web-based network management for remote control. Ideal for broadcast applications such as cable TV headends and secure broadcasting systems.

Key feature

Supports up to 1024 IP inputs via dual GE ports with optional SFP interface, max 840Mbps per port over UDP/RTP.

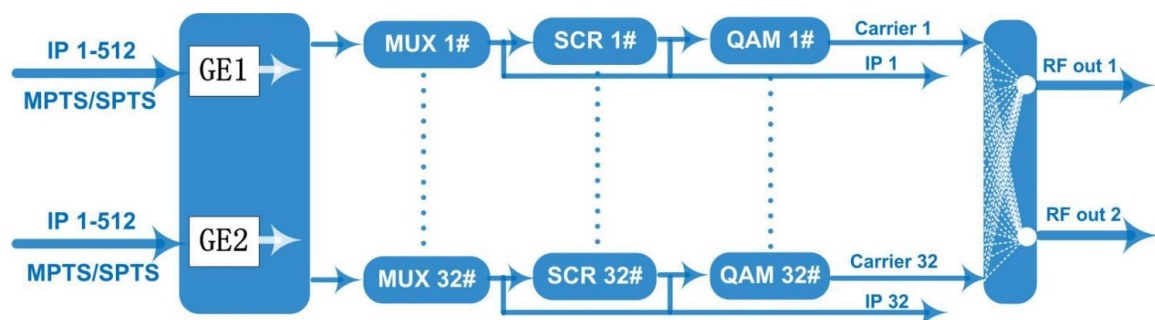
Outputs 32 non-adjacent QAM carriers via dual RF ports (50-960MHz), compliant with DVB-C (EN 300 429) and ITU-T J.83 A/B.

Integrates 32-channel multiplexing, scrambling, and modulation with PID remapping (up to 256 per channel), PCR adjusting, and PSI/SI editing.

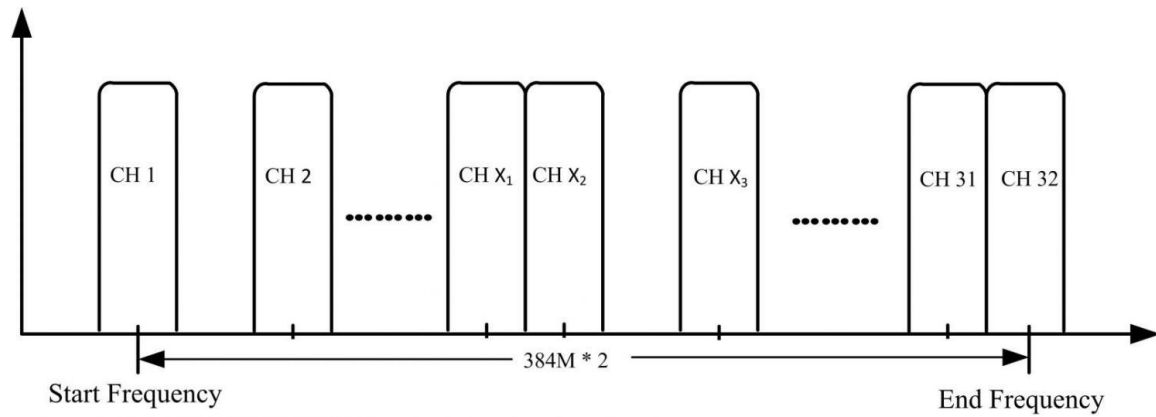
Complies with DVB scrambling standards including ETR289, ETSI 101 197, and ETSI 103 197.

Features web-based network management for remote control and monitoring.

Inner Principle Chart



Carrier Setting Illustration



Specification

Input	Input	Input	512×2 IP input, 2*100/1000M Ethernet Port(SFP interface optional)	512×2 IP input, 2*100/1000M Ethernet Port(SFP interface optional)
Input	Transport Protocol	Transport Protocol	TS over UDP/RTP, unicast and multicast, IGMP V2/V3	TS over UDP/RTP, unicast and multicast, IGMP V2/V3
Input	Transmission Rate	Transmission Rate	Max 840Mbps for each GE input	Max 840Mbps for each GE input
Mux	Input Channel	Input Channel	1024	1024
Mux	Output Channel	Output Channel	32	32
Mux	Max PIDs	Max PIDs	256 per channel	256 per channel
Mux	Functions	Functions	PID remapping(automatic/manual optional)	PID remapping(automatic/manual optional)
Mux	Functions	Functions	PCR accurate adjusting	PCR accurate adjusting
Mux	Functions	Functions	PSI/SI table automatically generating	PSI/SI table automatically generating
Scrambling Parameters	Max simulcrypt CA	Max simulcrypt CA	4	4

Scrambling Parameters	Scramble Standard	Scramble Standard	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197
Scrambling Parameters	Connection	Connection	Local/remote connection	Local/remote connection
Modulation Parameters	DVB-C Modulator Section	DVB-C Modulator Section	J.83A	Constellation :16/32/64/128/256QAM
Modulation Parameters	DVB-C Modulator Section	DVB-C Modulator Section	J.83A	Bandwidth :8M
Modulation Parameters	DVB-C Modulator Section	DVB-C Modulator Section	J.83B	Constellation :64/256QAM
Modulation Parameters	DVB-C Modulator Section	DVB-C Modulator Section	J.83B	Bandwidth :6M
Modulation Parameters	QAM Channel	QAM Channel	32 non-adjacent carrier	32 non-adjacent carrier
Modulation Parameters	Modulation Standard	Modulation Standard	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B
Modulation Parameters	Symbol Rate	Symbol Rate	5.0~7.0Msps, 1ksps stepping	5.0~7.0Msps, 1ksps stepping
Modulation Parameters	Constellation	Constellation	16, 32, 64 , 128, 256QAM	16, 32, 64 , 128, 256QAM
Modulation Parameters	FEC	FEC	RS (204, 188)	RS (204, 188)
RF Output	Interface	Interface	2 F type output ports for 32 carriers, 75Ω	2 F type output ports for 32 carriers, 75Ω
RF Output	RF Range	RF Range	50~960MHz, 1kHz stepping	50~960MHz, 1kHz stepping
RF Output	Output Level	Output Level	-20dBm~+10dBm(87~117dBμV), 0.1dB stepping	-20dBm~+10dBm(87~117dBμV), 0.1dB stepping
RF Output	MER	MER	≥ 40dB	≥ 40dB
RF Output	ACLR	ACLR	-60 dBc	-60 dBc

TS output	32 IP output over UDP/RTP/RTSP, unicast/multicast, 2*100/1000M Ethernet Ports (SFP)	32 IP output over UDP/RTP/RTSP, unicast/multicast, 2*100/1000M Ethernet Ports (SFP)	32 IP output over UDP/RTP/RTSP, unicast/multicast, 2*100/1000M Ethernet Ports (SFP)	32 IP output over UDP/RTP/RTSP, unicast/multicast, 2*100/1000M Ethernet Ports (SFP)
System	Network management software (NMS) supporting	Network management software (NMS) supporting	Network management software (NMS) supporting	Network management software (NMS) supporting
General	Demission	420mm×440mm×44.5mm (W×L×H)	420mm×440mm×44.5mm (W×L×H)	420mm×440mm×44.5mm (W×L×H)
General	Weight	3kg	3kg	3kg
General	Temperature	0~45■(operation), -20~80■(storage)	0~45■(operation), -20~80■(storage)	0~45■(operation), -20~80■(storage)
General	Power Supply	AC 100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz	AC 100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz	AC 100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz
General	Consumption	15.4W	15.4W	15.4W

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

Contact:Tomleequan Email■sales@fmuser.com whatsapp: [+8613922702227](https://wa.me/8613922702227)